

Polynomial Operations: Challenge Menu

Name: _____ Date: _____ Period: _____

Directions: Choose problems from each section. Show all work in the space provided.

SECTION A: Error Analysis (Identify the error, explain it, and give the correct answer.)

- A1.** Marcus simplified $(2x + 5)^2$ and got $4x^2 + 25$. Is Marcus correct? If not, explain his error and provide the correct answer.
- A2.** Keisha subtracted $(3x^2 - 4x + 7) - (x^2 + 2x - 5)$ and got $2x^2 - 2x + 2$. Is Keisha correct? If not, explain her error and provide the correct answer.
- A3.** David multiplied $-3x^2(2x^3 - 5x + 4)$ and got $-6x^5 + 15x - 12$. Is David correct? If not, explain his error and provide the correct answer.

SECTION B: Always, Sometimes, or Never True? (Circle your answer and justify.)

- B1.** The product of two binomials is a trinomial.

ALWAYS SOMETIMES NEVER

Justification:

- B2.** When you square a binomial, the middle term is positive.

ALWAYS SOMETIMES NEVER

Justification:

- B3.** Squaring a binomial results in a trinomial.

ALWAYS SOMETIMES NEVER

Justification:

- B4.** The sum of two polynomials has a greater degree than both of the original polynomials.

ALWAYS SOMETIMES NEVER

Justification:

SECTION C: Geometry Applications (*Write and simplify polynomial expressions.*)

C1. A rectangle has length $(4x + 3)$ and width $(2x - 5)$.

a) Write a simplified expression for the **perimeter** of the rectangle.

b) Write a simplified expression for the **area** of the rectangle.

C2. A square has side length $(3x - 2)$. A second square has side length $(x + 4)$. Write a simplified expression for the **difference** in their areas (larger minus smaller).

C3. A box has dimensions: length = $(x + 5)$, width = $(x - 1)$, and height = $(2x)$. Write a simplified expression for the **volume** of the box.

SECTION D: Fraction Operations (*Simplify completely. Show all work.*)

D1. $\left(\frac{1}{2}x^2 + \frac{2}{3}x - 1\right) + \left(\frac{3}{4}x^2 - \frac{1}{3}x + \frac{1}{2}\right)$

D3. $\frac{3}{4}x^2 \left(8x^3 - \frac{2}{3}x + 4\right)$

D2. $\left(\frac{2}{3}x^2 - \frac{1}{4}x + \frac{5}{6}\right) - \left(\frac{1}{6}x^2 + \frac{3}{4}x - \frac{1}{3}\right)$

SECTION E: Work Backwards (*Find the missing polynomial or term.*)

E1. What polynomial must be added to $(3x^2 - 5x + 2)$ to get $(7x^2 - 2x - 4)$?

E2. Find A and B: $(Ax + 3)(3x + B) = 6x^2 + 13x + 6$